Cross-Basin Generalization for Tropical Cyclone Forecasting using TropiCycloneNet

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Problem Context

Tropical cyclone (TC) forecasting in data-scarce ocean basins is limited by small training sets. The South Pacific (SP) contains 30 storms in the TropiCycloneNet dataset, compared to 131 in the Western Pacific (WP), a ratio that makes it impractical to train reliable models on SP alone [1]. Transfer learning from data-rich to data-scarce basins is a possible solution, and there is preliminary evidence this works: Qu et al. (2025) fine-tuned a Western North Pacific-trained transformer on the Eastern North Pacific and outperformed operational forecasting systems [2].

We focus on WP to SP as the transfer pair because it presents a structural challenge beyond sample size. WP sits in the Northern Hemisphere and SP sits in the Southern Hemisphere. The Coriolis force acts in opposite directions in each hemisphere, so WP storms tend to move northwest/west while SP storms tend to move southeast/west/southwest. A model trained on WP will therefore systematically mispredict SP storm directions. However, the atmospheric physics governing TC intensity, specifically sea-surface temperature and vertical wind shear, is shared across both basins. DeMaria and Kaplan (1994) identify these as key intensity predictors [3], and our exploratory analysis confirms that their distributions overlap substantially between WP and SP (Fig. 4).

Proposed Approach

We evaluate five architectures on zero-shot WP to SP transfer, then fine-tune the best performers. Each architecture offers a different reason to expect cross-basin generalisation:

Model	Why it might transfer
U-Net	Strong spatial feature extraction from 3D weather data (wind, temperature, pressure) and a well-established baseline for weather prediction
ResNet	Learns hierarchical features from gridded weather data with deep residual connections that help capture complex atmospheric patterns
FNO	Operates in frequency space so learned patterns are not tied to a specific grid or basin
PI-GAN	Designed to enforce physical conservation laws (vorticity, potential vorticity) that hold across hemispheres
Transformer	Fuses all three data modalities (1D, 3D, environmental) through cross-attention, allowing flexible reweighting across basins

Three-stage pipeline:

1. *Zero-shot*: Train each model on WP, remap directional labels to account for hemisphere differences, evaluate on SP with no fine-tuning.
2. *Fine-tune*: Adapt top-performing models on 10/25/50% of SP data, measure how accuracy improves with increasing amounts of SP data.
3. *Extension*: Test whether the best models generalise to other data-scarce basins (NI, SI) without basin-specific retraining.

Exploratory Data Analysis

Four findings from our preliminary analysis support the proposed WP to SP transfer. The Basin Similarity Matrix in Fig. 1 shows that WP and SP have the lowest average Wasserstein distance across physical variables of any cross-hemisphere basin pair. Storm tracks in Fig. 2 show clear geographic separation and hemisphere differences in movement. Direction distributions in Fig. 3 confirm that WP and SP movement patterns are hemisphere-mirrored, with WP storms predominantly moving northwest and west while SP storms predominantly move southeast and west. This can be addressed by remapping directional labels before transfer. The vertical wind shear distributions in Fig. 4 overlap substantially between WP and SP, supporting the argument that intensity-related physics is shared across both basins.

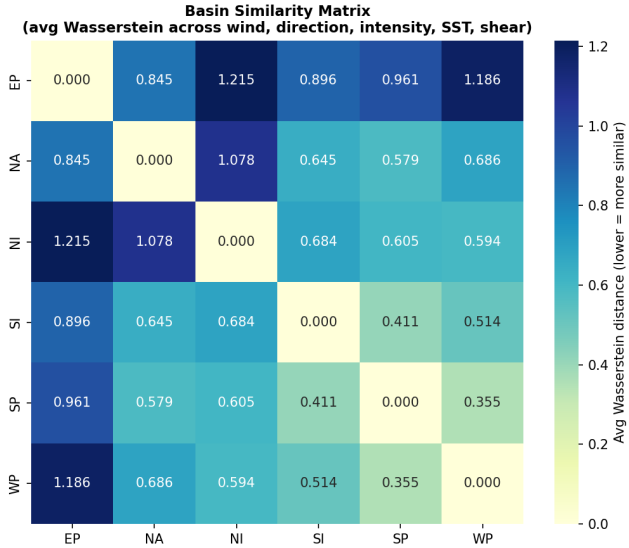

Fig. 1: Average Wasserstein distance across wind, direction, intensity, SST and shear for all basin pairs. Lower is more similar.

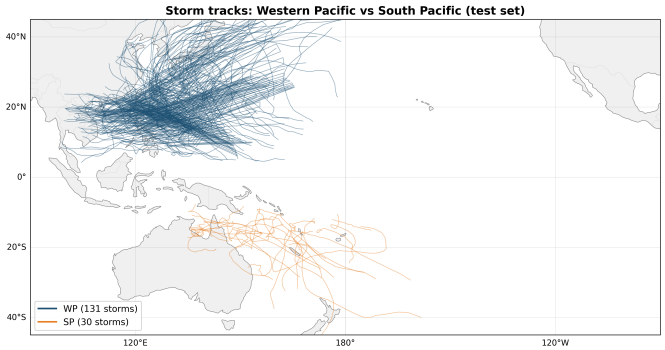

Fig. 2: WP (131 storms) and SP (30 storms) storm tracks.

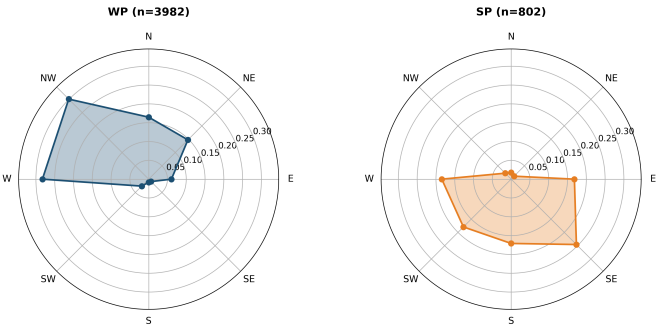

Fig. 3: Direction polar roses for WP (n=3982) and SP (n=802).

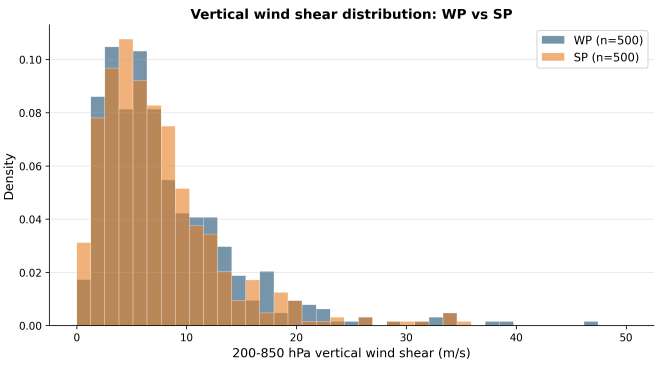

Fig. 4: Vertical wind shear distributions for WP and SP (n=500 each).

References [1] Huang et al., *Nat. Commun.*, 2025. [2] Qu et al., *npj Clim. Atmos. Sci.*, 2025. [3] DeMaria & Kaplan, *Wea. and Forecasting*, 1994. [4] Knapp et al., *Bull. Amer. Meteor. Soc.*, 2010.